

AUTHENTICATION

HTTP method	Route	Description	Role required	Request body	Expected Response
POST	/api/auth/register	Creates a new user account, either an organiser or a participant.	None	{ fullName, email, password, role }	201 Created – returns new user id and role. 400 Bad Request – validation failed or email already registered.
POST	/api/auth/login	Authenticates a user and issues an access token.	None	{ Email, password }	200 OK – RETURNS JWT and user role. 401 Unauthorized – invalid credentials.

User Profile

HTTP method	Route	Description	Role required	Request body	Expected Response
GET	/api/users/{id}	Retrieves a users profile details	Any(own profile)	None	200 OK – user profile object. 403 forbidden – requesting another users profile. 404 Not Found – user does not exist.
PUT	/api/users/{id}	Updates a users own profile information.	Any(own profile)	{ fullName, email }	200 OK – user profile object. 403 forbidden – not the profile. 404 Not Found – user does not exist.
GET	/api/users/{id}/results	Returns a participants full personal performance history across all events entered.	Any(own profile)	None	200 OK – array of result records. 403 forbidden – not the profile owner.

Events

HTTP method	Route	Description	Role required	Request body	Expected Response
GET	/api/events	List all upcoming events for participants to browse.	None	None	200 OK – array of events.
GET	/api/events/{id}	Retrieves full details for a single event, including its route and weather – relevant location data.	None	None	200 OK – event object. 404 Not Found – event does not exist.
POST	/api/events	Creates a new event	Organiser	{ eventName, eventDate, location, description }	201 Created – returns new event id. 400 Bad Request – Validation failed.
PUT	/api/events/{id}	Updates an existing events details	Organiser	{ eventName, eventDate, location, description }	200 Ok – updated event. 403 Forbidden – not the events organiser. 404 Not Found – event does not exist.
DELETE	/api/events/{id}	Removes an event and its dependent categories, routes and enrolments.	Organiser	None	204 No Content. 403 forbidden – not the events organiser. 404 Not Found – event does not exist
GET	/api/events/{id}/route	Returns the route and map data for an event so participants can prepare for race day.	None	None	200 OK – route object. 404 Not Found – no route captured for this event.

Categories

HTTP method	Route	Description	Role required	Request body	Expected Response
GET	/api/events/{id}/categories	Lists all entry categories available for an event	None	None	200 OK – array of categories
POST	/api/events/{id}/categories	Adds a new category to an event	Organiser(owner only)	{ CategoryName, distanceKm, maxParticipants }	201 Created – returns new category id. 403 Forbidden – not the events organiser.
PUT	/api/categories/{id}	Updates and existing category	Organiser	{ CategoryName, distanceKm, maxParticipants }	200 OK – updated category. 403 Forbidden – not the events organiser. 404 Not Found – category does not exist
DELETE	/api/categories/{id}	Removes a category and its dependent enrolments	Organiser	None	204 No Content. 403 Forbidden – not the events organiser.

Event Enrolments

HTTP method	Route	Description	Role required	Request body	Expected Response
POST	/api/categories/{id}/enrolments	Enters the logged in participant into a category, subject to the categorys participant limit	Participant	{ } (participant identified from token)	201 Created – returns enrolment id and status. 404 Not Found – category does not exist. 409 Conflict – category is full or participant already enrolled.
GET	/api/users/{id}/enrolments	Lists all of a participants current and past enrolments	Any (own enrolments only)	None	200 OK array of enrolments. 403 Forbidden – not the profile owner.
DELETE	/api/enrolments/{id}	Cancels an enrolment before the event takes place.	Participant(own enrolment or organiser)	None	204 No Content. 403 Forbidden – not the enrolment owner or event organiser. 404 Not Found – enrolment does not exist

Results

HTTP method	Route	Description	Role required	Request body	Expected Response
POST	/api/enrolments/{id}/results	Captures a finish time and position for a participants enrolment after race day	Organiser (event owner only)	{ finishTime, position }	201 Created – returns new result id. 403 Forbidden – not the events organiser. 404 Not Found – enrolment does not exist. 409 Conflict – result already captured.
GET	/api/events/{id}/results	Returns the full public results lists for an event, across all categories.	None	None	200 OK array of results. 404 Not Found – event does not exist.
GET	/api/categories/{id}/results	Returns the public results list for a single category, ranked by finish time	None	None	200 OK array of results ordered by position. 404 Not Found – category does not exist.